

12-Month SDE Roadmap (Internship + Job Prep)

This roadmap is designed for Computer Science students targeting internships in <6 months and full-time SDE placements in 12–15 months. The journey is divided into two parts: **Part A (Months 1–6):** Internship Preparation → Focus on DSA fundamentals + Web Dev basics + projects. **Part B (Months 7–12):** Job Preparation → Advanced DSA, System Design, Full-stack project, and optional ML/AI projects.

Part A: Internship Preparation (Months 1–6)

Month	Focus Areas	Expected Output
1	DSA: Arrays, Strings, Hashing Web: HTML, CSS, JavaScript basics	Solve 30–40 DSA problems, Build simple static pages
2	DSA: Recursion, Linked List Web: DOM, ES6 JS, Events	40–50 problems, To-do List project
3	DSA: Stacks, Queues, Binary Trees Web: React basics	40 problems, Portfolio site
4	DSA: Binary Search, Sorting, Searching Web: React components, Hooks	30 problems, Weather/Quiz App
5	DSA: Basic DP, Graphs intro Web: Node.js, Express basics	30 problems, Blog backend
6	DSA: Mixed practice + mock tests Web: MongoDB, integrate MERN mini project	Complete 150–200 problems, Deploy 2–3 projects on GitHub

Part B: Job Preparation (Months 7–12)

Month	Focus Areas	Expected Output
7	DSA: Advanced DP Web: Authentication (JWT/OAuth)	30 problems, Add login to projects
8	DSA: Graphs (BFS, DFS, Shortest Paths) Web: Full-stack app integration	30 problems, Deploy MERN app
9	DSA: Tries, Heaps, Priority Queues System Design (LLD basics)	30 problems, Learn 2–3 design patterns
10	DSA: Segment Trees, Binary Indexed Trees System Design (HLD basics)	30 problems, Simple system design (URL Shortener)
11	DSA: Company-specific problem sets Web: Collaborative full-stack project	40 problems, Team project on GitHub
12	DSA: Revise + Mock interviews Optional: ML mini projects (Spam filter, Recommender)	400+ problems total, 1–2 ML projects for resume

Daily Routine Suggestion:

- 2–3 hrs → DSA practice (LeetCode, GFG, Striver's Sheet)
- 1.5–2 hrs → Web development (learning + projects)
- Weekends → Resume polishing, GitHub updates, optional ML projects